

			= 13.6 m [A1]	
6	(b)	(ii)	1. no, no [A1] 2. no, halved [A1]	
7	(a)	(i)	$80 - 45 = 35 \text{ m}$	A1
7	(a)	(ii)	$\pi (45)^2 = 6360 \text{ m}^2$	A1
7	(a)	(iii)	blade tip speed = 7×10 $T = 2\pi r/v = 2\pi(45)/(7 \times 10) = 4.0 \text{ s}$	C1 A1
7	(b)	(i)	either output power $\geq 3 \text{ MW}$ for wind speed between 14 and 25 m s^{-1} so yes or output power $< 3 \text{ MW}$ for wind speed below 14 m s^{-1} so no	A1
7	(b)	(ii)	power consumption per household = $470 \text{ kW h} = 470\,000 / (30 \times 24)$ = 653 W [accept 29-31] number of homes = $3\,000\,000 / 653 = 4600$	C1 A1
7	(c)	(i)	1. 3.2 MW 2. 15 m s^{-1} (or 14 to 16)	A1 A1
7	(c)	(ii)	1. $E = \frac{1}{2} \pi (45)^2 (15)^3 (1.25)$ = $1.34 \times 10^7 \text{ W}$ 2. $3\,200\,000 / 1.34 \times 10^7 \times 100\%$ = 24% (23.9%)	C1 A1 C1 A1
7	(c)	(iii)	$1.34 \times 10^7 \times (100 - 24 - 10)\%$ = $8.84 \times 10^6 \text{ W}$	C1 A1
7	(c)	(iv)	1. 25 m s^{-1}	A1

			2. (The generators are designed to take a certain maximum electrical load.) Having a cut-out speed will prevent the circuit from being overloaded and damaged.	A1
7	(d)	(i)	tip of rotor blade [B1] because it is a sharp point (and lightning tends to strike sharp points) [B1]	B1 B1
7	(d)	(ii)	install grounding down conductors (or lightning rods) in the blades (or use poor electrical conductor such as fiberglass for blades.)	A1